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# Q19 - Image Acquisition and Computer Vision Features for Self-Driving Cars
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import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# Upload Image
# -----
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Image Acquisition
# -----
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)

# -----
# Noise Reduction
# -----
blur = cv2.GaussianBlur(gray, (5,5), 0)

# -----
# Edge Detection
# -----
edges = cv2.Canny(blur, 50, 150)

# -----
# Detect Objects using Contours
# -----
contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL,
                               cv2.CHAIN_APPROX_SIMPLE)

detected = img.copy()

for cnt in contours:
    if cv2.contourArea(cnt) > 500:
        x, y, w, h = cv2.boundingRect(cnt)
        cv2.rectangle(detected, (x, y), (x+w, y+h),
                       (255,0,0), 2)

# -----
# Display Results
# -----
plt.figure(figsize=(16,10))

plt.subplot(2,2,1)
plt.imshow(img)
plt.title("Original Image (Image Acquisition)")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(2,2,3)
plt.imshow(edges, cmap="gray")
plt.title("Detected Edges")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(detected)
plt.title("Detected Features / Objects")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# Information
# -----
print("Computer Vision Pipeline")
print("-----")
print("#1 - Camera acquires the road image.")
```

```

print("1. Camera acquires the road image.")
print("2. Image is converted to grayscale for easier processing.")
print("3. Gaussian Blur removes noise.")
print("4. Canny Edge Detection extracts road boundaries.")
print("5. Contours identify possible vehicles, signs, or obstacles.")
print("6. These visual features help the self-driving system")
print("    make safe driving decisions such as lane keeping,")
print("    obstacle avoidance, and braking.")

```

Choose files 00000161-0...2ef0000.avif

00000161-0097-d257-a169-c2b7a2ef0000.avif(image/avif) - 67161 bytes, last modified: 14/07/2026 - 100% done
 Saving 00000161-0097-d257-a169-c2b7a2ef0000.avif to 00000161-0097-d257-a169-c2b7a2ef0000.avif

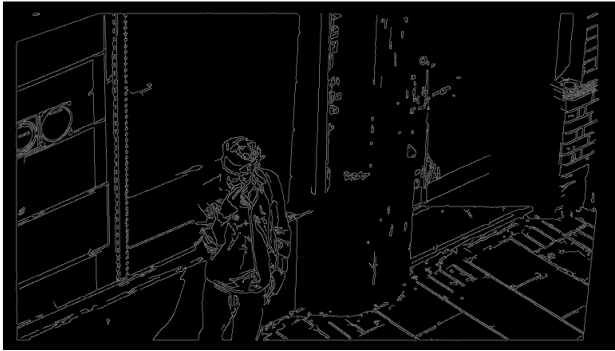
Original Image (Image Acquisition)



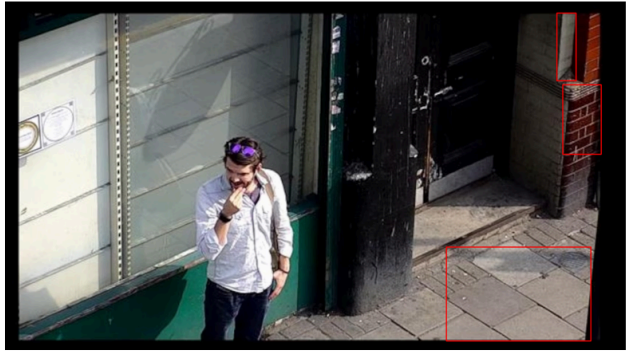
Grayscale Image



Detected Edges



Detected Features / Objects



Computer Vision Pipeline

1. Camera acquires the road image.
2. Image is converted to grayscale for easier processing.
3. Gaussian Blur removes noise.
4. Canny Edge Detection extracts road boundaries.
5. Contours identify possible vehicles, signs, or obstacles.
6. These visual features help the self-driving system make safe driving decisions such as lane keeping, obstacle avoidance, and braking.

